Odmocniny

1. Usměrněte zlomky:

$a)\ \dfrac{\sqrt{3}-\sqrt{2}}{\sqrt{2}}$

$b)\ \dfrac{4-\sqrt{3}}{\sqrt{3}+4}$

2. Částečně odmocněte:

$\sqrt{50}$

$\sqrt{72}$

3. $(9\sqrt{2}+4\sqrt{3})(9\sqrt{2}-4\sqrt{3}) = (9\sqrt{2})^2-(4\sqrt{3})^2 = 81\cdot 2-16\cdot 3 =$

$A+B \quad A-B \qquad A^2-B^2$

4. $\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}+\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} = \dfrac{(\sqrt{5}-\sqrt{3})^2+(\sqrt{5}+\sqrt{3})^2}{(\sqrt{5}+\sqrt{3})(\sqrt{5}-\sqrt{3})} = \dfrac{5-2\sqrt{15}+3+5+2\sqrt{15}+3}{5-3}$

5. $\dfrac{\sqrt{a^{\frac{1}{3}}a}}{\sqrt[3]{(\sqrt{a}.a^{-2})^2}} = \dfrac{(a^{\frac{1}{3}}a)^{\frac{1}{2}}}{\left[(a^{\frac{1}{2}}\cdot a^{-2})\right]^{\frac{2}{3}}} = \dfrac{\left(a^{\frac{4}{3}}\right)^{\frac{1}{2}}}{\left(a^{-\frac{3}{2}}\right)^{\frac{2}{3}}} = \dfrac{a^{\frac{2}{3}}}{a^{-1}} = a^{\frac{2}{3}+1} = a^{\frac{5}{3}} = a^{1\frac{2}{3}} = a\sqrt[3]{a^2}$

6. $\sqrt{x\sqrt[3]{x\sqrt[4]{x}}} = \left[x\left(x\cdot x^{\frac{1}{4}}\right)^{\frac{1}{3}}\right]^{\frac{1}{2}} = \left[x\cdot\left(x^{\frac{5}{4}}\right)^{\frac{1}{3}}\right]^{\frac{1}{2}} = \left[x\cdot x^{\frac{5}{12}}\right]^{\frac{1}{2}} = \left[x^{\frac{17}{12}}\right]^{\frac{1}{2}} = x^{\frac{17}{24}} = \sqrt[24]{x^{17}}$

10. $\dfrac{x^{\frac{5}{12}}.y^{\frac{5}{6}}.y^{-\frac{1}{2}}}{x^{-\frac{3}{4}}.y^{\frac{1}{3}}.x^{\frac{2}{3}}} = x^{\frac{5}{12}-\left(-\frac{3}{4}+\frac{2}{3}\right)}\ y^{\frac{5}{6}-\frac{1}{2}-\frac{1}{3}}$

11. $\left(a^{\frac{3}{2}}:\ a^{-\frac{1}{3}}\right):\left(a^{\frac{5}{3}}.\ a^{\frac{1}{2}}\right)$

12. $\left[2.\left(2\,.\,2^{\frac{1}{2}}\right)^{\frac{1}{2}}\right]^{\frac{1}{2}}$

13. $2^{\frac{1}{2}}.4^{\frac{1}{3}}.8^{\frac{1}{4}}.16^{\frac{1}{6}}.32^{\frac{1}{12}} = 2^{\frac{1}{2}}\cdot\left(2^{2}\right)^{\frac{1}{3}}\cdot\ \ldots$

14. $\sqrt[5]{\frac{4}{\sqrt[3]{2}}:\sqrt[3]{\frac{2}{\sqrt[5]{8}}}} = \left[\frac{2^{2}}{2^{\frac{1}{3}}}:\left(\frac{2}{2^{\frac{3}{5}}}\right)^{\frac{1}{3}}\right]^{\frac{1}{5}} = \left[2^{2-\frac{1}{3}}:2^{\frac{1}{3}-\frac{1}{5}}\right]^{\frac{1}{5}}$

$2^{\frac{3}{5}\cdot\frac{1}{3}}$

15. $\left[\left(2^{2}\cdot\frac{1}{3}\right)^{2}\right]^{\frac{1}{3}}:\left[\left(\frac{1}{2}.3^{2}\right)^{\frac{1}{3}}\right]^{2} = \left[\left(2^{2}\cdot 3^{-1}\right)^{2}\right]^{\frac{1}{3}}:\left[\left(2^{-1}\cdot 3^{2}\right)^{\frac{1}{3}}\right]^{2} = \left[\left(2^{2}\right)^{\frac{2}{3}}\cdot 3^{-\frac{2}{3}}\right]:\left[2^{-\frac{2}{3}}\cdot\left(3^{2}\right)^{\frac{2}{3}}\right]$

$= \left[2^{\frac{4}{3}}\cdot 3^{-\frac{2}{3}}\right]:\left[2^{-\frac{2}{3}}\cdot 3^{\frac{4}{3}}\right] = 2^{\frac{4}{3}+\frac{2}{3}}\cdot 3^{-\frac{2}{3}-\frac{4}{3}} = 2^{2}\cdot 3^{-2} = \frac{4}{9}$

16. $\left(\sqrt[3]{\frac{x\sqrt{x}}{x^{-2}}}:\sqrt{\frac{x^{-3}\sqrt{x}}{x^{2}}}\right)^{-1}\cdot\frac{x^{-3}\sqrt{x}}{\sqrt[3]{x^{2}}} = \left[\left(\frac{x\cdot x^{\frac{1}{2}}}{x^{-2}}\right)^{\frac{1}{3}}:\left(\frac{x^{-3}\cdot x^{\frac{1}{2}}}{x^{2}}\right)^{\frac{1}{2}}\right]^{-1}\cdot\frac{x^{-3}\,x^{\frac{1}{2}}}{x^{\frac{2}{3}}} =$